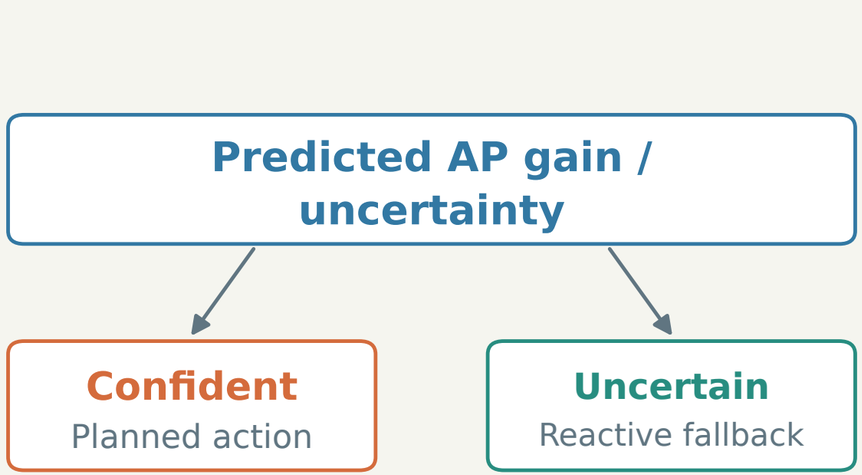
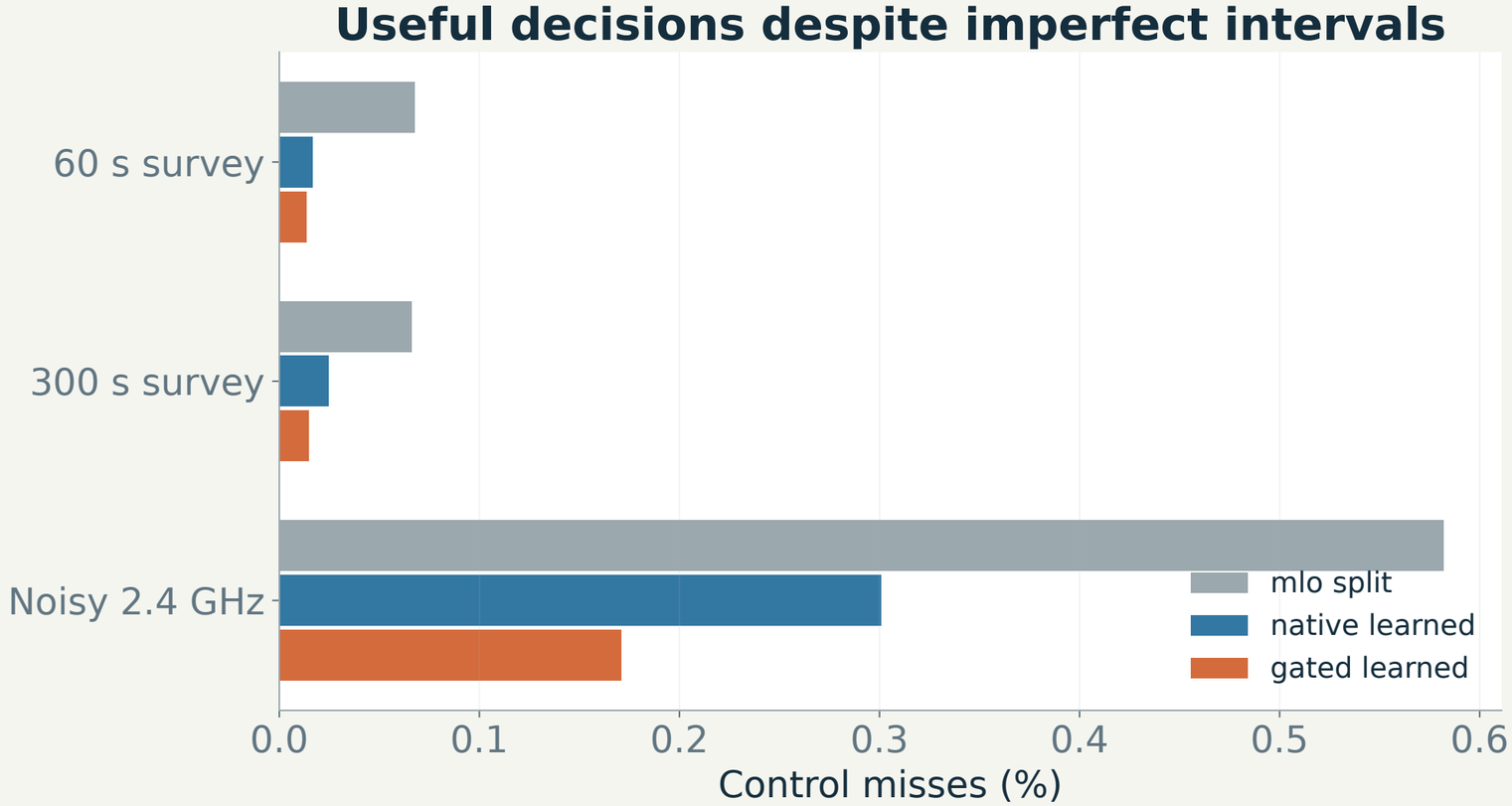
$$z = \Delta / \sqrt{\sigma_{old}^2 + \sigma_{new}^2}$$